

Is there a difference in means between Alpine and Mercedes Pitstops?

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Introduction

Data collected from every race of the Formula 1, at each venue the pitstop time for every team has been recorded. With this, we aim to provide a decision as to whether there is a difference in means between Alpine and Mercedes pitstops and what find out what the population mean is. Providing analysis to data in F1 is vital since cars can be split by thousandths of a second, therefore by identifying the average time for pitstop in F1 can give teams valuable information as to whether they are under or overperforming. Similarly, a comparison between Mercedes and Alpine will give a direct comparison in pitstop times, and will tell us whether there is statistical evidence, to answer the question of which team is faster in the pits.

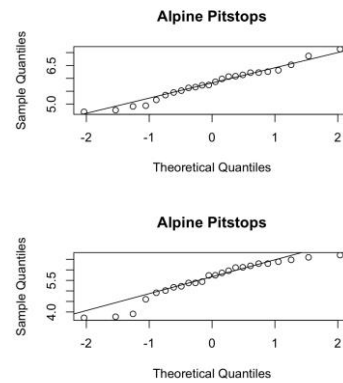
Exploratory Analysis

Table 1: Sample means from Alpine, Mercedes and all the teams.

Mean Pitstop (seconds, 3 d.p.)		
<i>All Teams</i>	<i>Mercedes</i>	<i>Alpine</i>
5.703	5.802	5.542

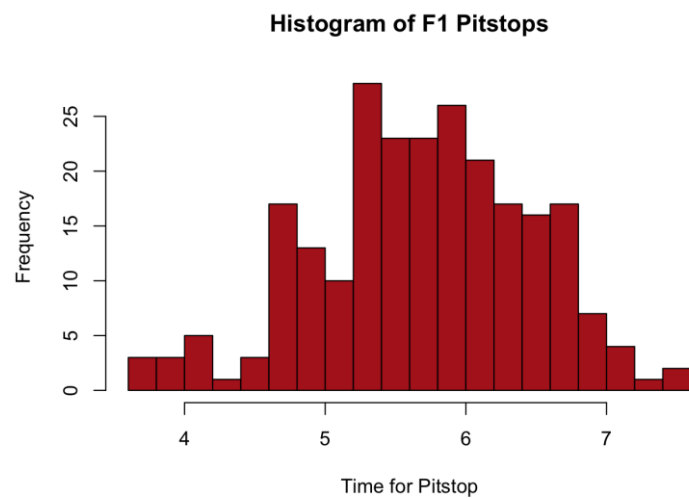
The initial impression from Table 1 is that Alpine make faster pitstops, however with a difference of 0.260 seconds it is not clear that the difference is significant.

Figure 2: QQ plots for the data



The QQ plots shows it does follow a normal distribution and so we could model as so.

Figure 1: Histograms showing the distribution of all pitstops in the F1 season.



Looking at Figure 1, pitstops across the season for all F1 teams seem to follow a normal distribution too. Further it looks symmetric and unimodal with no extreme outliers.

Statistical Analysis

To justify a reason for a difference between the average pitstops at Mercedes and Alpine we will calculate the 99% confidence interval for the difference in means. From the information provided in Figure 2 we can see that the yields of all the distributions approximately follow a normal distribution. We need to calculate the 99% confidence interval with the following criteria; an unknown, but equal standard deviation. The formula used is as follows:

$$CI_{1-\alpha}(\mu_X - \mu_Y) = \left[(\bar{x} - \bar{y}) - t_{1-\frac{\alpha}{2}, \nu_p} \cdot s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}}, \right. \\ \left. (\bar{x} - \bar{y}) + t_{1-\frac{\alpha}{2}, \nu_p} \cdot s_p \sqrt{\frac{1}{n_X} + \frac{1}{n_Y}} \right] \quad (8.15)$$

Using this, we find that the 99% confidence interval for the difference in mean pitstops between Mercedes and Alpine to be $(-0.333, 0.853)$. Since 0 is still included in this interval we cannot say that there is sufficient evidence to say that there is a difference in means between Mercedes and Alpine pitstops.

Now we will investigate the population mean pitstop time, we will calculate a 99% interval with a given $\sigma = 0.8$, then with an unknown standard deviation. We will model the mean as $\bar{X} \rightarrow N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$, since we established earlier that the pitstops approximately follow a normal distribution, and so we will find the interval easily from this. The interval computed from this is, $(5.570, 5.837)$, so we can be 99% confident that the population mean lies between these two values.

Finally, we re-calculate the 99% confidence interval with an unknown standard deviation.

Given $\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$, we can then use the formula:

$$\bar{X} \pm t_{n-1, 0.995} \cdot \frac{s}{\sqrt{n}}$$

Computing with that gives us an interval of, $(5.576, 5.831)$, and so, using this technique we can be 99% confident that the mean pitstop lies in between these two values.

Conclusion

Now that we have completed our statistical analysis, it can be said that we have answered our aims. In our exploratory analysis we saw from Table 1 that on average Alpine had slightly faster pitstops compared to Mercedes, however it's been discussed that there is no strong statistical evidence to suggest a difference in the mean pitstop times between Mercedes and Alpine since 0 was included in the interval for the difference of means. Also, our initial thought from Figure 2 was that the model approximately followed a normal distribution, and so it allowed us to construct two 99% confidence interval, one with a given known $\sigma = 0.8$ and the other with an unknown σ . After computation we found them to be (5.570, 5.837) and (5.576, 5.831) respectively, both of which are very narrow intervals with extremely similar values, and so we can say that the true mean for pitstops across all the F1 teams should be around these two intervals. Overall, we find that our data does not indicate a difference in the average pitstop between Mercedes and Alpine and that the average pitstop for all teams is within a narrow range.